

Bayesian multimodeling: graphical models

2025

Graphical models

Conditional independence

Events X, Y are conditionally independent w.r.t. Z : $X \perp Y|Z$, if

$$P(X|Y, Z) = P(X|Z).$$

Conditional dependence

Events X, Y are conditionally dependent w.r.t. $\mathcal{G}$: $X, Y \in \mathcal{G}$, if

$$X \not\perp Y | \mathcal{G} \setminus \{X, Y\}.$$

Graphical models

A probability model is graphical, if it can be represented as a graph, where the edges correspond to conditionally dependent events.

Non-graphical models

- MLP, decision trees, etc.
- Models with complex behaviour:

$$Y = X_1 * X_2 * X_3 + X_1 * X_2 + X_1 * X_3 + X_2 * X_3$$

Types of graphical models

- Directed models (aka Bayesian networks)
 - ▶ Easy to design
- Undirected (Markov models)
- Factor-graphs
 - ▶ Easy to infer and optimize

Plate notation

Plate notation is an alternative visualization for graphical models.

Elements:

- White circles (random variables);
- Grey circles (observed variables);
- Small circles (deterministic values);
- Plates (batching).

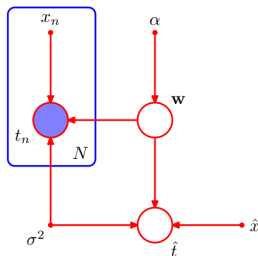


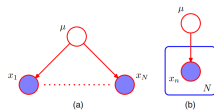
Plate notation for linear regression (Bishop)

Bayesian networks

- Models are set using directed acyclic graphs
- Joint distribution for the graph with K vertices:

$$p(v_1, \dots, v_K) = \prod_{i=1}^K p(v_i | \text{parent}(v_i))$$

- Example: linear regression



DAG and Plate notation (Bishop)

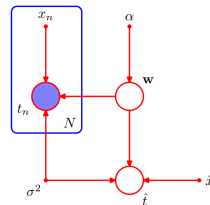


Plate notation for regression model (Bishop)

Causality graph elements

$$X \rightarrow Y \rightarrow Z - \text{chain}$$

Example:

- X — school budget
- Y — average student score
- Z — university acceptance ratio

Properties:

- ① X and Y , Y and Z are dependent:
 $\exists x, y : \mathbf{P}(Y = y | X = x) \neq p(Y = y)$
 $\exists y, z : \mathbf{P}(Z = z | Y = y) \neq p(Z = z)$
- ② Z and X : are (probably) dependent
- ③ $Z \perp X | Y$: are conditionally independent: $\forall x, y, z$

$$\mathbf{P}(Z = z | X = x, Y = y) = \mathbf{P}(Z = z | Y = y)$$

(if Y is fixed, then X and Z are independent)

Causality graph elements

$$X \leftarrow Y \rightarrow Z \text{ — fork}$$

Example:

- X — ice cream sells
- Y — average temperature
- Z — crime ratio

Properties:

- ① X and Y , Y and Z are dependent
- ② X and Z are (probably) dependent
- ③ $X \perp Z | Y$ are conditionally independent

Causality graph elements

$$Y \rightarrow X \leftarrow Z \text{ — collider}$$

Example (illness):

- X — entered MIT
- Y — math skills
- Z — engineering skills

Properties:

- ① Y and X , Z and X are dependent
- ② Y and Z are independent
- ③ $Y \not\perp Z | X$ are conditionally dependent

d-separation

The path P is **blocked** by Z , if:

- ① P contains $A \rightarrow B \rightarrow C$, $A \leftarrow B \rightarrow C$, $B \in Z$
- ② P contains $A \rightarrow B \leftarrow C$, $B \notin Z$ and all children of $B \notin Z$

If Z blocks all the paths from X to Y , then X and Y are **d-separated**:

$$X \perp Y | Z.$$

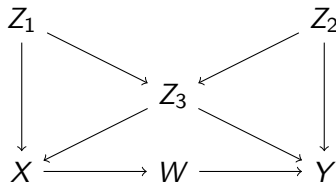
d-separation

The path P is blocked by Z , if:

- 1 P contains $A \rightarrow B \rightarrow C$, $A \leftarrow B \rightarrow C$, $B \in Z$
- 2 P contains $A \rightarrow B \leftarrow C$, $B \notin Z$ and all children of $B \notin Z$

If Z blocks all the paths from X to Y , then X and Y are d-separated.

Example:



Pair	d-separation set
(Z_1, W)	X

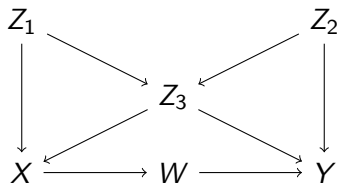
d-separation

The path P is blocked by Z , if:

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If Z blocks all the paths from X to Y , then X and Y are d-separated.

Example:



Pair	d-separation set
(Z_1, W)	X
(Z_1, Y)	$\{Z_3, X, Z_2\}, \{Z_3, W, Z_2\}$

Model selection for Bayesian networks

- Generally, NP-hard problem
- Reduces to optimization problem with predefined search space or sampling problem
- Independence determination:
 - ▶ ML and MAP
 - ▶ Evidence
 - ▶ Information criteria

Markov random fields

Models are represented as undirected graphs.

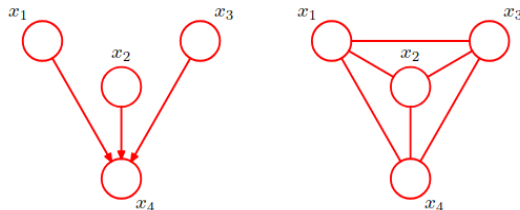
Difference from Bayesian networks:

- No direction $\rightarrow$ cannot infer causality.
- The likelihood is factorized as follows:

$$p(\mathbf{x}) = \frac{1}{Z} \prod_C \psi(\mathbf{x}_C),$$

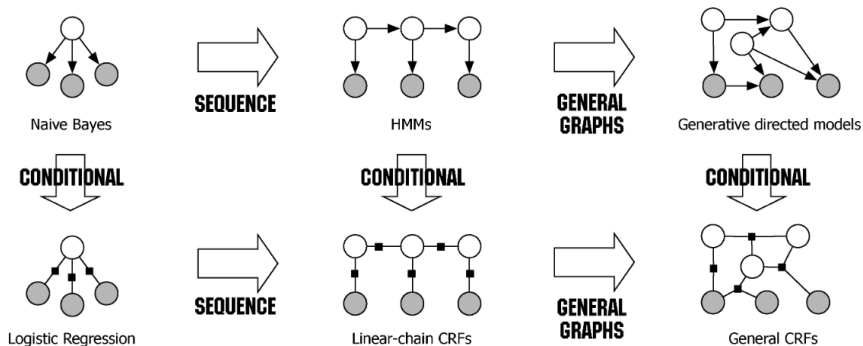
where $\mathbf{x}_C$ is a maximal clique, $\psi \geq 0$ is a potential function.

- Conditional independence: if all the paths from A to B go through C , then $A \perp B | C$.



(Bishop)

Example: CRF and HMM



(Sutton, McCallum)

Inference in chains



(Bishop)

Naive likelihood calculation for x_n :

$$p(x_n) = \sum_{x_1} \sum_{x_2} \cdots \sum_{x_{n-1}} \sum_{x_{n+1}} \cdots \sum_{x_N} p(\mathbf{x}),$$

For N discrete variables with K values the complexity is $O(K^N)$

Inference in chains: regrouping

$$p(x_n) = \sum_{x_1} \sum_{x_2} \dots, \sum_{x_{n-1}} \sum_{x_{n+1}} \dots \sum_{x_N} p(\mathbf{x}),$$

$$p(\mathbf{x}) = \psi(x_1, x_2) \psi(x_2, x_3) \dots \psi(x_{N-1}, x_N).$$

Regroup the sum:

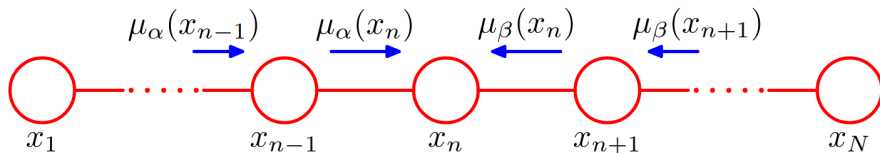
$$\begin{aligned} p(x_n) = & \sum_{x_{n-1}} \psi(x_{n-1}, x_n) \dots \left(\sum_{x_1} \psi(x_1, x_2) \right) \times \\ & \times \left(\sum_{x_{n+1}} \psi(x_n, x_{n+1}) \dots \left(\sum_{x_N} \psi(x_{N-1}, x_N) \right) \right). \end{aligned}$$

Now complexity is $O(NK^2)$.

Message passing

$$p(x_n) = \underbrace{\sum_{x_{n-1}} \psi(x_{n-1}, x_n) \dots \left(\sum_{x_1} \psi(x_1, x_2) \right)}_{\mu_a(x_n)} \times \underbrace{\left(\sum_{x_{n+1}} \psi(x_n, x_{n+1}) \dots \left(\sum_{x_N} \psi(x_{N-1}, x_N) \right) \right)}_{\mu_b(x_n)}.$$

Interpretation: $\mu_a(x_n)$ is a message transferred from x_{n-1} to x_n , $\mu_b(x_n)$ is a backward message from x_{n+1} .



Inference in chains: details

The inference is iterative:

- calculate $\sum_{x_1} \psi(x_1, x_2) = \mu_a(x_2)$, that stores $\mu_a(x_2)$ for each value of x_2 ;
- calculate $\sum_{x_2} \psi(x_2, x_3) (\sum_{x_1} \psi(x_1, x_2)) = \sum_{x_2} \psi(x_2, x_3) \mu_a(x_2) = \mu_a(x_3)$;
- ...
- calculate $\sum_{x_{n+1}} \psi(x_n, x_{n+1}) \mu_b(x_{n+1}) = \mu_b(x_n)$.
- for directed variables, where

$$\psi(x_1, x_2) = p(x_1)p(x_2|x_1), \quad \psi(x_i, x_{i+1}) = p(x_{i+1}|x_i),$$

μ_b should not be calculated:

$$\begin{aligned} \mu_b(x_n) &= \sum_{x_{n+1}} \psi(x_n, x_{n+1}) \dots \left(\sum_{x_N} \psi(x_{N-1}, x_N) \right) = \\ &= \sum_{x_{n+1}} p(x_{n+1}|x_n) \dots \left(\sum_{x_N} p(x_N|x_{N-1}) \right) = 1. \end{aligned}$$

Factor graph

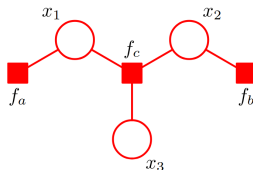
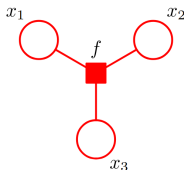
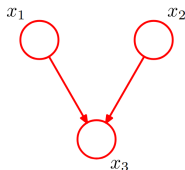
Definition

Factor-graph is a bipartite graph with two types of vertices: variables and factors.
The likelihood is a production of factors:

$$p(\mathbf{x}) = \prod_i f_i.$$

Example: model $p(x_1)p(x_2)p(x_3|x_2, x_1)$ has two variants of factorization:

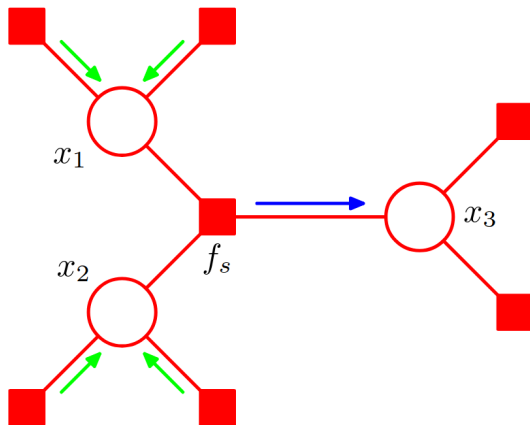
$$f = p(x_1)p(x_2)p(x_3|x_2, x_1), \quad f_a = p(x_1), f_b = p(x_2), f_c = p(x_1)p(x_2)p(x_3|x_2, x_1).$$



(Bishop)

Inference in factor-graphs: example

Sum-product: likelihood is a composition of messages from factors to variables.



Model examples: RBM

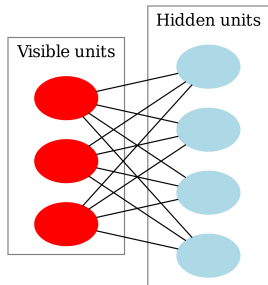
$$p(\mathbf{x}, \mathbf{h}) = \frac{1}{Z} \exp(-E(\mathbf{x}, \mathbf{h})),$$

$$E = -\mathbf{w}_1^\top \mathbf{x} - \mathbf{w}_2^\top \mathbf{h} - \mathbf{x}^\top \mathbf{W}_3 \mathbf{h},$$

$p(\mathbf{h} = 1|\mathbf{x})$ and $p(\mathbf{x} = 1|\mathbf{h})$ are 1-layers with sigmoid activation.

Derivative of the log-likelihood for a single example $\mathbf{x}$ with respect to $\mathbf{w}$:

$$\frac{\partial \log p_{\mathbf{w}}(\mathbf{x})}{\partial \mathbf{w}} = \mathbb{E}_{\mathbf{x}', \mathbf{h} \sim p_{\mathbf{w}}(\mathbf{x})} \left[\frac{\partial E_{\mathbf{w}}(\mathbf{x}')}{\partial \mathbf{w}} \right] - \frac{\partial E_{\mathbf{w}}(\mathbf{x})}{\partial \mathbf{w}}.$$



Model examples: Structured VAEs

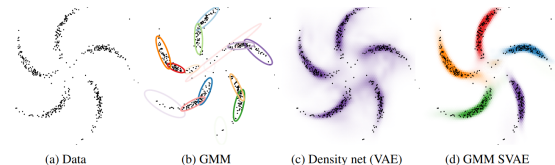
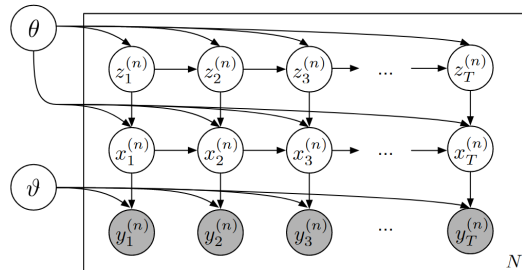
Based on SLDS:

$$z_{t+1}|z_t \sim \pi^{t+1},$$

$$\mathbf{y}_t \sim \mathcal{N}(\text{MLP}^{z_t}(\mathbf{x}_t)).$$

Optimization: optimize ELBO.

Inference: message-passing.



What happens with Graphical models now

- Standard in causal inference
- GNN: uses message passing from GM
- Combination of DL and GM: see previous slide
- See also pyro and other frameworks

References

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- Sutton, C, McCallum, A 2012, 'An Introduction to Conditional Random Fields', Foundations and Trends in Machine Learning, vol. 4, no. 4, pp. 267-373
- <https://www.stat.umn.edu/geyer/5421/notes/graph.pdf>

Organizational issues

- Checkpoint: I will start checking tomorrow
- General requirements for library and algorithms:
 - ▶ Codestyle: at least PEP-8
 - ▶ The code must be commented
 - ▶ There must be an installation file (setup.py or alternative)
 - ▶ There must be requirements file with defined versions
 - ▶ All these points (and general work of library) will be checked offline
- Demo: must be available at the meeting (perfect case: ipynb or collab)
- Tests:
 - ▶ Coverage 75%;
 - ▶ Tests during build: not required
 - ▶ Will be checked offline
- Documentation: must be ready and deployed
- Blog-post and reports: pre-final version must be ready
- Documentation, report and blog-post will be checked offline. For blog-post and tech report prepare for the external review.